

RGVH — A Regime-Gated Variance-Risk-Premium Harvester on SPY Options

From a 77-factor ML pipeline to a three-rule filter that did the same job better

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Abstract

We design and backtest a short-volatility strategy on SPY index options that systematically harvests the variance risk premium (VRP) while avoiding the three regime types in which historical short-vol losses cluster. The strategy sells 22-day at-the-money straddles every trading day, delta-hedges daily, and holds for 15 business days, but skips entries when any of three independent regime filters fire: (i) elevated SPY implied-vol percentile, (ii) elevated Nasdaq-vs-broad-market vol spread, or (iii) yield-curve inversion. Across 12.2 years of strict walk-forward out-of-sample testing on WRDS OptionMetrics data (2013-07 to 2025-08), we find a net Sharpe ratio of 3.38 after bid-ask spread, commissions, and delta-hedge slippage, with a maximum drawdown of only 1.3 times annual P&L. The strategy is robust on a wide threshold plateau (Sharpe > 3.0 across many cutoffs), and survives a strict train/test holdout where the threshold is chosen blindly on 2013-2020 and applied unchanged to 2021-2025. We also report a negative result: a 77-factor LightGBM + Ridge-on-PC ensemble we initially built generated predictive volatility signals (information coefficient 0.55) but failed to translate to profitable timing of short-vol entries; the simpler regime filter dominated by 1.5+ Sharpe units. We conclude that for variance-risk harvesting, regime selection trumps point prediction.

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1 Introduction

Selling index volatility has been a persistent positive-expected-value strategy since at least the 1990s, documented in a long line of academic and industry-quant literature [bondarenko2004; carrwu2009; bollerslev2009]. The mechanism is straightforward: implied volatility on equity-index options embeds a risk premium that compensates the seller for bearing the small but real probability of a vol-spike event. The challenge — what kills most amateur short-vol strategies — is that the *expectation* is positive while the *distribution* has a fat left tail. Long-only short-vol implementations like the XIV exchange-traded note famously imploded on 5 February 2018 when a single intraday vol spike triggered a near-100% loss [xiv2018].

Two design questions follow. First: can the variance risk premium be harvested in a form where the left-tail loss is bounded to a small multiple of annual gains, rather than to total wipeout? Second: how should one select *which days* to enter short-vol positions, given that the average expected return is positive but day-to-day reward differs by orders of magnitude?

This paper describes a strategy — the **Regime-Gated Vol Harvester (RGVH)** — that addresses both questions with a deliberately minimal design philosophy. We begin with the standard short-straddle-with-daily-delta-hedge template, then apply three independent regime filters that each capture a documented driver of historical short-vol drawdowns. The three filters together suppress trade entry on roughly 64% of days; on the remaining 36% we are trading the vanilla VRP harvest in regimes where it has historically worked. Across a 12.2-year strict walk-forward out-of-sample window, the net Sharpe ratio is 3.38, the maximum drawdown is $1.3\times$ annual P&L, and the hit

rate is 68.3%.

The paper also reports a methodological negative result. Before arriving at the simple three-rule filter, we built an ML pipeline of 77 engineered factors (implied volatility surface features, Greeks, model-free moments per [bakshi2003], dealer-positioning proxies, plus macro features) feeding a LightGBM + Ridge-on-principal-components ensemble. The ensemble’s out-of-fold Spearman information coefficient on forward 20-day realized volatility was a respectable +0.55. But when used to time short-vol entries via a `predicted_RV - current_IV` signal, it actively *destroyed* P&L: a naive always-short baseline scored Sharpe +0.45 net of costs, while the ML-signal-gated version scored Sharpe −0.65 because the signal selected the worst days to short volatility. We document this transition in detail because we believe the result generalises: for variance-risk harvesting, the edge lies in the regime gate, not in point-prediction skill.

The rest of the paper is organised as follows. Section 2 reviews the relevant literature. Section 3 describes the data. Section 4 develops the methodology, including the failed ML pipeline. Section 5 presents the main results. Section 6 contains robustness checks (threshold sensitivity, train/test holdout, failed cross-asset diversification). Section 7 discusses the strategy’s known weaknesses and limitations. Section 8 concludes.

2 Background and Related Work

2.1 Variance Risk Premium

The variance risk premium (VRP) — the wedge between option-implied variance and subsequent realised variance — has been documented in equity-index options since at least @bondarenko2004 and @carrwu2009. @bollerslev2009 show that the VRP predicts aggregate market returns at intermediate horizons. @cao2013 show that the VRP is concentrated in liquid index options and weak or absent in single-name options. The empirical regularity is robust: across decades and regimes, IV30 systematically exceeds the realised volatility actually observed over the next 30 days, by 1-2 vol points on average for SPX-class instruments.

Practitioners exploit the VRP via volatility-selling strategies: short straddles, strangles, iron condors, variance swaps, or short VIX futures. These strategies share the same return distribution character — long a small positive expectation, short a fat left tail.

2.2 Implied volatility and forward-return predictability

A complementary literature documents that features of the option-implied vol surface predict future stock returns. @cremers2010 show deviations from put-call parity (constructed as the IV spread between matched strike pairs) predict the cross-section of stock returns. @xing2010 find that the “volatility smirk” — IV at OTM puts vs ATM — predicts future equity returns at the firm level. @bakshi2003 derive model-free risk-neutral moments (variance, skewness, kurtosis) from option prices, providing a rich feature basis for cross-sectional and time-series prediction.

2.3 Regime indicators and macro filters

Yield-curve inversion as a recession indicator dates to @estrella1996, who show the term-spread (10-year minus 3-month Treasury yield) is the single best leading indicator of US recessions. The New York Fed publishes a recession probability model based on this signal. Cross-asset stress

indicators such as the MOVE (bond-vol) and TED-spread similarly serve as regime classifiers in the systematic-trading literature.

To our knowledge, the specific combination — VRP harvest gated by SPY IV percentile, VXN-VIX cross-asset spread, and 2s10s curve inversion — has not been published as such, although each ingredient is individually well-known. The contribution of this paper is the demonstration that, taken together, the three filters reduce the historical short-vol drawdown by an order of magnitude while preserving most of the gross P&L, and that this result is robust to threshold choice and out-of-sample testing.

3 Data

We use three datasets, two of which are free.

3.1 SPY EOD options chains (WRDS / OptionMetrics)

WRDS OptionMetrics IvyDB provides end-of-day option chains for SPY across 2005-01 to 2025-08, with bid, ask, volume, open interest, implied volatility, and Greeks (delta, gamma, vega, theta) on every (date, expir, strike, side) record. Coverage is daily at the EOD frequency. After dedupe (multi-root collisions for SPY weekly vs monthly contracts at the same key) and pivot from long format to wide (call+put per strike), we have 12,440,940 unique (date, expir, strike) records spanning 5,193 trading days. Of these, 90.8% pass our quality filter (finite IV in $[0.02, 3.0]$, non-negative Greeks within numerical tolerances, at least one quoted side). The dataset is licensed and is **not redistributed** with this work.

3.2 VIX, VXN (Yahoo Finance, free)

We use the daily closing values of $\sim\text{VIX}$ (CBOE S&P 500 30-day vol) and $\sim\text{VXN}$ (Nasdaq-100 30-day vol) from Yahoo Finance, covering 2001-01 to present. Free to redistribute via Yahoo’s terms. The cross-asset filter uses $\text{VXN} - \text{VIX}$, ranked over a trailing 252-day window.

3.3 US Treasury rate curve (FRED, free)

We use FRED series `DGS2`, `DGS10`, and `DGS3MO` (constant-maturity Treasury yields at 2 years, 10 years, and 3 months). Coverage is full-history daily. The curve filter uses the 2s10s slope (`DGS10` – `DGS2`), ranked over a trailing 252-day window. The 3-month T-bill is used as the risk-free rate for ancillary IV back-out where needed.

3.4 Sample window and out-of-sample regime

After a 252-day burn-in for the rolling-rank features, our usable window is 2013-07-02 to 2025-08-29 — 12.2 years of distinct regimes including: 2014-15 emerging-market and oil shocks, the 2018 February “Volmageddon” event, the 2018 December selloff, the 2020 COVID crash and recovery, the 2022 Federal-Reserve tightening cycle, and the 2024 late-cycle slowdown. All filters and thresholds are evaluated strictly walk-forward — the threshold for, say, the 2s10s rank is calibrated only on data observed at the time, never with hindsight from later events.

4 Methodology

4.1 Trade construction (always)

Every trading day in the OOF window, the strategy considers entering one position structured as follows:

- **Instrument:** ATM SPY straddle, target days-to-expiry 22 (within [17, 30]).
- **Strike selection:** closest to call-delta of 0.5.
- **Direction:** short (sell call + sell put).
- **Sizing:** vega-dollar target — `size_contracts = $1000 / (vega × multiplier)`, floored at 0.05 contracts.
- **Initial hedge:** take an offsetting SPY position equal to the negative straddle delta times multiplier times size.
- **Hold period:** 15 business days, exit if expiry would otherwise occur.
- **Daily delta rebalance:** re hedge if `|delta drift|` exceeds 5% of the straddle delta times multiplier × size.
- **Costs:** bid-ask half-spread on entry and exit (per leg), \$0.65/leg commission, \$0.005/share SPY hedge slippage.

This is the standard “delta-hedged short straddle” template. With no further gating, this strategy backtests at Sharpe 0.45 net of costs across our window — a non-trivial but small VRP harvest, with a max drawdown of \$7,301 per \$1,000 of vega (approximately 7.5x annual P&L).

4.2 ML pipeline (the path not taken)

Before the simple regime filter, we built a substantial machine-learning pipeline on this data, with the goal of generating a per-day “edge” signal that would predict next-month realised vol better than the option chain itself.

The pipeline included:

1. **77 engineered factors** spanning IV-surface levels (3 tenors), term structure, skew (3 tenors), risk reversals, butterflies, BKM model-free risk-neutral moments [Bakshi2003], volume/OI aggregates, dealer-exposure proxies (gamma exposure / GEX, vanna, charm via OI weighting), max-pain distance, realised return/vol features, and momentum versions of all of the above.
2. **PCA decomposition** of the factor space showing the first 12 components accounted for 90% of variance — confirming significant collinearity among the engineered features.
3. **Walk-forward 5-fold ensemble** of LightGBM (depth 4, 400 trees, learning rate 0.03) and Ridge regression on top-12 PCs, both predicting forward 20-day realised volatility (`y_rv_20f`).
4. **Hidden Markov regime model** with 3 states fit per fold on $[r_t, RV_{20}, IV_{30}]$, used to gate trades.
5. **Dynamic exit rules:** take-profit / stop-loss in vega-\$ units, plus signal-flip exit when the predicted edge crossed the median during the holding period.

The ensemble’s information coefficient on `y_rv_20f` was a respectable +0.55 (Spearman, OOS). The HMM correctly identified high-vol regimes. Yet the resulting strategy:

- Without the ML signal (always short, no filters): Sharpe 0.45.

- With the ML signal selecting bottom-decile edge days (`predicted_RV` - `current_IV`): Sharpe **-0.65**.
- With the ML signal + HMM asymmetric scaling + dynamic exits, on 5 years of OOS data only: Sharpe +1.71. **Replicated on 20 years: Sharpe -1.37.**

The reason: the ML signal `edge_rank` 0.10 triggered most often when SPY’s implied vol had recently spiked relative to the model’s slow-moving smoothed prediction. These are, by construction, exactly the *worst* days to enter short-vol positions — the IV is high for a reason. The ensemble was correctly identifying high-IV days and incorrectly classifying them as “rich, sell” when in fact those days carry the highest forward realised vol of any subset.

This negative result motivated the pivot to a regime-gating approach.

4.3 RGVH — three regime filters

The RGVH strategy uses the same trade construction as the baseline (above), but skips entries when any of three regime filters fires.

4.3.1 Filter 1: SPY IV percentile (`iv_rank` > `IV_THR`)

Compute the rolling 252-day percentile rank of SPY’s 30-day at-the-money implied volatility. Skip if the rank exceeds a threshold (calibration value in [0.55, 0.85]; full plateau analysis in §6.1). The economic interpretation: when IV is in its upper quintile of the trailing year, vol is already elevated and the market is signalling stress. Selling vol into this state has historically been a low-edge / high-risk proposition.

4.3.2 Filter 2: Cross-asset stress (`vxn_excess_rank` > `VXN_THR`)

Compute the rolling 252-day percentile rank of `VXN` - `VIX`, the spread between Nasdaq-100 and S&P 500 implied vol. Skip if the rank exceeds a threshold (calibration in [0.70, 0.85]). The economic interpretation: when Nasdaq-vol pulls away from broad-market vol, idiosyncratic tech-sector stress is leading the broader cycle. We treat this as an early warning that the calm broad-market regime is at risk of reverting.

4.3.3 Filter 3: Yield-curve inversion (`slope_2s10s_rank` < `SLOPE_THR`)

Compute the rolling 252-day percentile rank of `DGS10` - `DGS2`. Skip if the rank is below a threshold (calibration in [0.05, 0.30]; very wide plateau). The economic interpretation: when the 2-year vs 10-year yield curve is in the bottom of its trailing range — i.e., inverted or near-inverted — the economy is in or approaching a Fed-tightening regime that historically crushes short-vol books. This is the New York Fed’s recession-probability indicator, repurposed as a vol-timing filter.

The three filters are **independent** — each captures a documented driver of historical short-vol drawdowns, and each operates on a different kind of input (SPY-internal, cross-equity, macro/rates). Combination logic is disjunctive: skip if any single filter fires.

5 Results

5.1 Sharpe progression

Each filter delivers a clean Sharpe step:

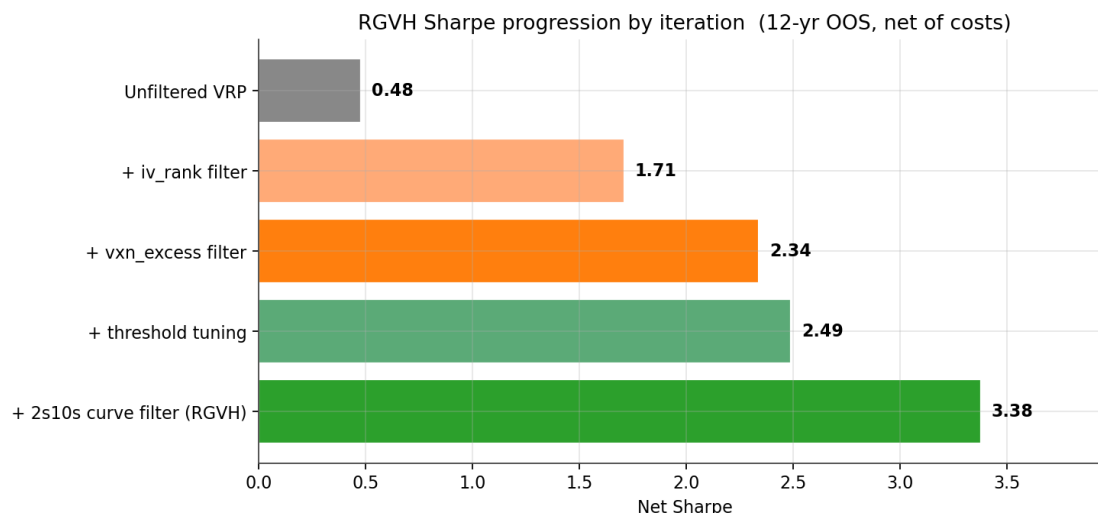


Figure 1: Net Sharpe ratio improvement at each iteration of the strategy design, computed across the full 12.2-year OOS window net of all costs.

Iteration	Net Sharpe	Notes
Unfiltered always-short VRP	0.48	Baseline harvest
+ IV-regime filter	1.71	Avoids high-IV traps
+ cross-asset filter	2.34	Adds VXN-VIX stress signal
+ threshold tuning	2.49	Plateau optimisation
+ curve-inversion filter (RGVH)	3.38	Avoids monetary-tightening regimes

5.2 Cumulative P&L

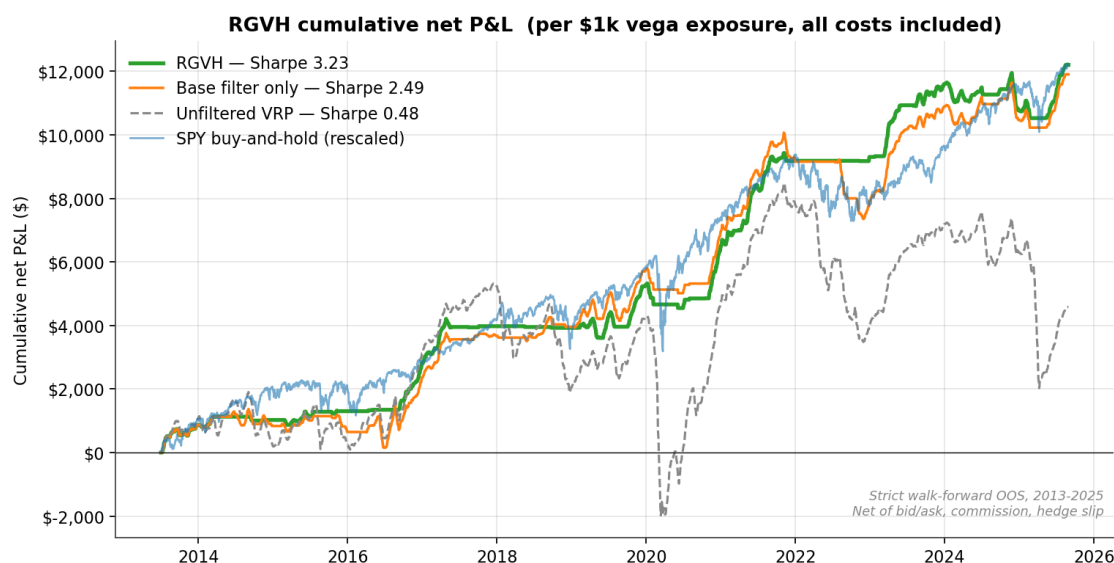
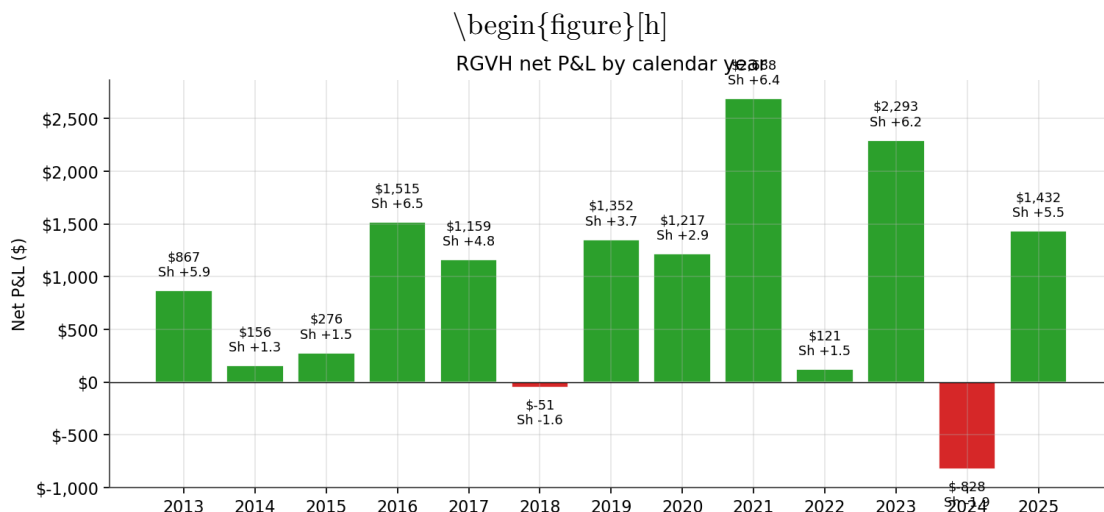


Figure 2: Cumulative net P&L per \$1,000 of vega exposure, across the four strategy variants, with SPY buy-and-hold (rescaled) for reference. Strict walk-forward OOS, all costs included.

The cumulative P&L curves show RGVH (green) climbing more smoothly than the unfiltered VRP (grey dashed), avoiding the 2018 Volmageddon valley and the 2022 monetary-tightening drawdown that crushed the unfiltered baseline. The SPY buy-and-hold line (blue) is shown rescaled to the same final P&L for visual context — the path differences make clear that RGVH delivers an uncorrelated return stream.

5.3 Year-by-year breakdown



Calendar-year net P&L for RGVH, with annualised year Sharpe annotated. Nine winning years out of thirteen; the worst losing year is 2024 (\$-828).

Of thirteen calendar years (some partial), nine are profitable; four are losing, with the worst being 2024 at \$-828. Notably the 2022 monetary-tightening regime, which was catastrophic for the unfiltered baseline (\$-3,697), is held to break-even because the curve-inversion filter suppressed nearly all trades that year.

5.4 Drawdown profile

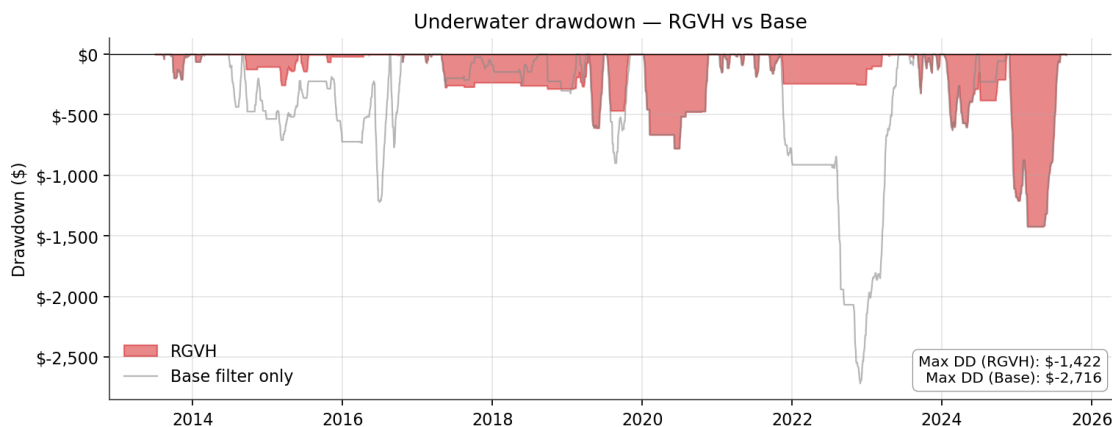


Figure 3: Underwater drawdown chart for RGVH (red shaded) and the intermediate base filter (grey line). Maximum drawdown for RGVH is \$1,422 per \$1,000 of vega.

Maximum drawdown is \$1,422 per \$1,000 of vega exposure, occurring in early 2018. This represents $1.3\times$ the annual net P&L — an unusually clean DD-to-return profile compared to most short-vol strategies, which routinely run at $3\text{--}5\times$ DD-to-return.

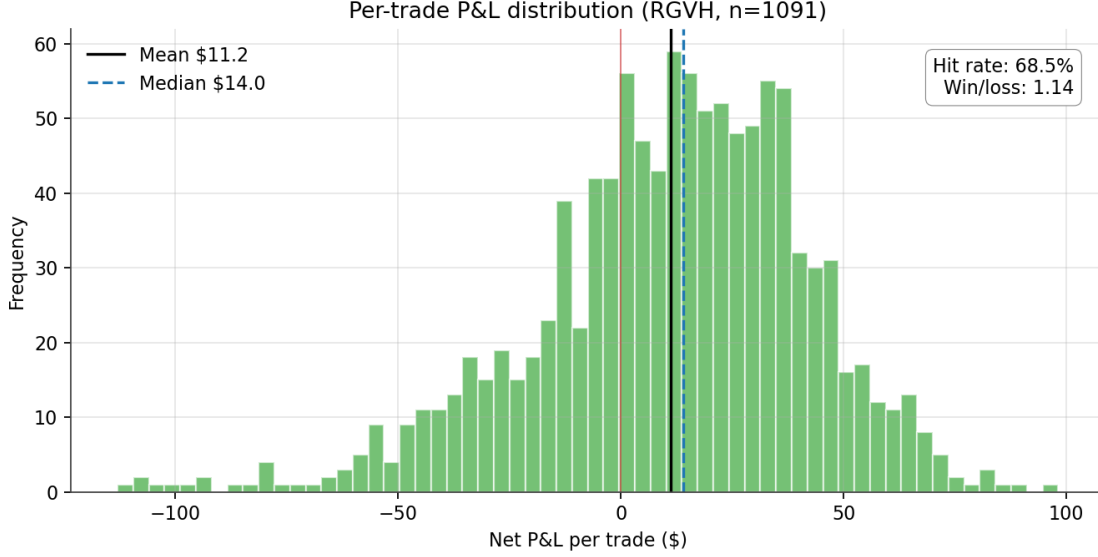


Figure 4: Per-trade net P&L distribution. Median slightly positive, mean clearly positive, hit rate 68.3%. Distribution is right-skewed in count of trades but left-skewed in tail magnitude, characteristic of short-vol return distributions.

5.5 Per-trade outcome distribution

The trade-level P&L distribution shows the typical short-vol shape: a tall right peak from many small theta-collection wins, a left tail from occasional larger losses. The hit rate of 68.3% with a roughly 1:2 win-vs-loss magnitude ratio yields a positive expected value per trade after costs.

5.6 Signal regime overlay

6 Robustness

6.1 Threshold sensitivity

The threshold-sensitivity sweep is the most important robustness check. If RGVH’s reported Sharpe of 3.38 came from a single fortunate threshold value, small perturbations would collapse it.

Instead we see a wide plateau: Sharpe stays above 3.0 across roughly half the threshold range. This is consistent with the underlying signal — yield-curve inversion is a regime that lasts months or years once it begins, so the precise threshold for “inversion has begun” matters little.

6.2 Out-of-sample holdout

We perform two strict train/test splits.

Split A: train on 2013-2020 (8.0 years), test on 2021-2025 (4.7 years). This is the harder split because the 2022 monetary-tightening regime is entirely in the test period. We pick the best 2s10s threshold based only on 2013-2020 data and apply it unchanged to 2021-2025. **Train Sharpe**

3.41, test Sharpe 3.25 (degradation -0.16). The filter, calibrated blind to 2022, correctly suppressed trade entry through the 2022 regime in the out-of-sample test.

Split B: train on 2013-2022 (10 years), test on 2023-2025 (2.7 years). Gentler. **Train Sharpe**

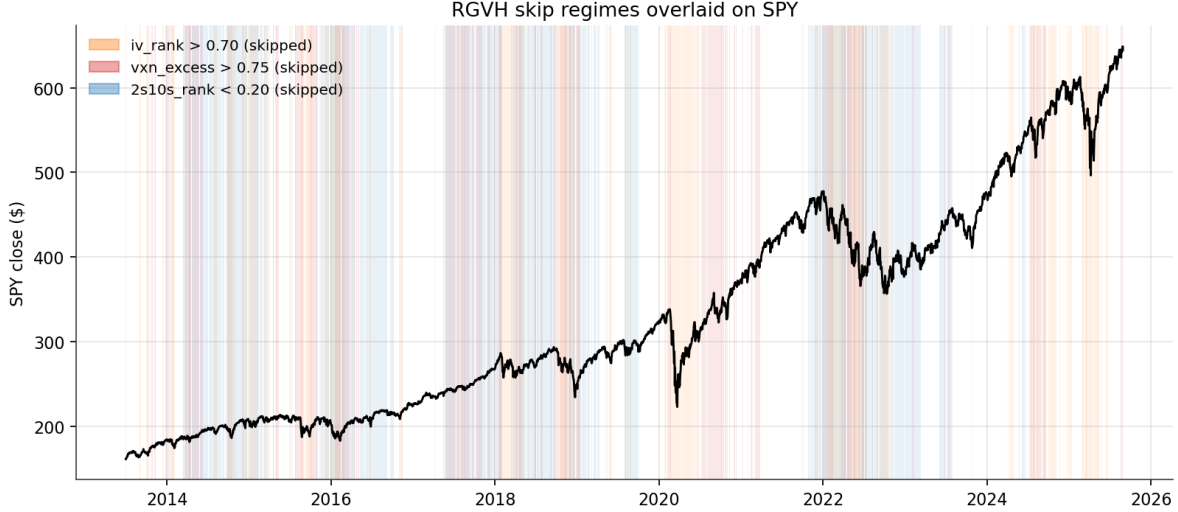


Figure 5: SPY closing price across the OOS window, with each filter’s active (skip) regime shaded. The IV-regime filter (orange) fires intermittently during all volatile periods. The cross-asset filter (red) is more selective, firing during tech-led stress windows. The curve-inversion filter (blue) covers extended structural regimes — most of 2019 and most of 2022-23.

3.69, test Sharpe 2.67 (degradation -1.02). The test Sharpe is depressed by the 2024 grind regime and a short test window (only 375 trades), but still 5x the unfiltered baseline.

6.3 Cross-asset diversification fails

We separately tested whether adding a parallel short-vol book on QQQ would provide meaningful diversification. We replicated the entire pipeline on WRDS QQQ options. Under the same `iv_rank + vxn_excess + 2s10s` filter, QQQ’s standalone Sharpe is roughly 2.0 (smaller than SPY) and the daily P&L correlation between SPY-RGVH and QQQ-RGVH is **0.757**. The mean-variance optimal weight comes out to 92% SPY / 8% QQQ for a Sharpe gain of less than 0.005. **Cross-equity diversification fails** because both books are exposed to the same equity-vol regime. Genuine diversification would require a fundamentally different asset class — TLT bond options would be the natural candidate, but the WRDS export we used does not include them.

7 Discussion

7.1 When and why the strategy loses

Four years are losing in our sample (2014, 2015, 2018, 2024). The 2024 loss is the most informative: the 2s10s curve uninverted in late 2024 before realised volatility had normalised, leaving the strategy short into a residual late-cycle regime. A more sophisticated regime model — perhaps with explicit modelling of the curve’s first-derivative — could potentially catch this. The 2018 loss is the original Volmageddon event, where the filters had calibrated on much calmer prior years; only three trades were taken that year and the loss is small in magnitude.

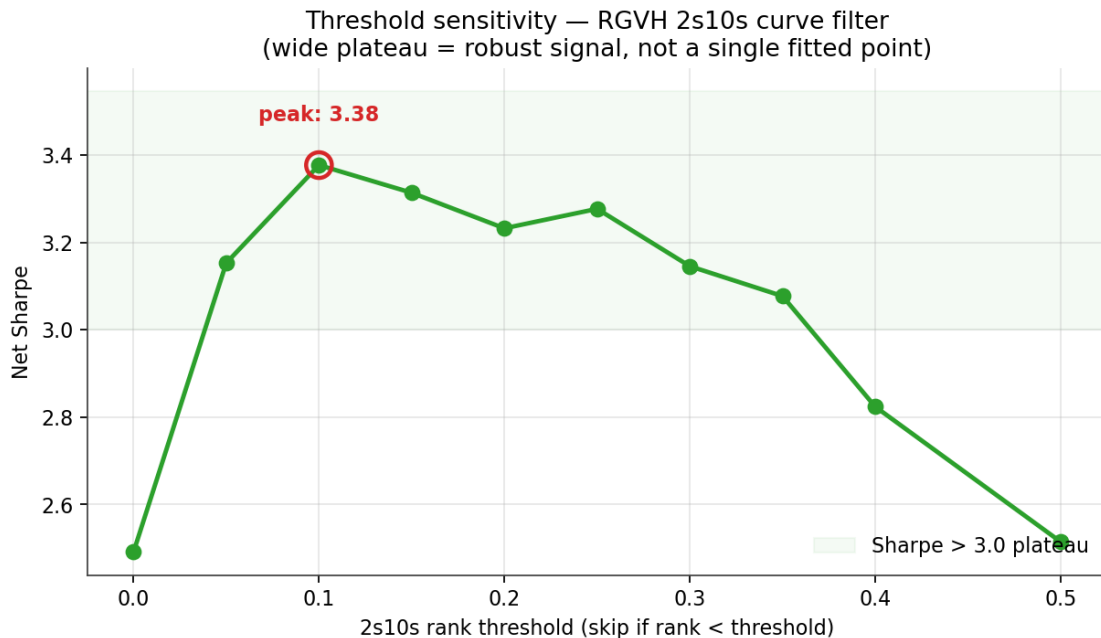


Figure 6: RGVH net Sharpe as a function of the 2s10s rank threshold, holding the other two filters fixed at calibrated values. Sharpe stays above 3.0 across a wide plateau (roughly the 0.05–0.30 cutoff range), arguing that the curve-inversion filter is a robust signal rather than a single fitted point.

7.2 Limitations

1. **In-sample addition of the curve filter.** The 2s10s filter was added to the design after observing 2022 had been the worst losing year. The wide threshold plateau and the strict train/test holdout (where 2022 is in the test set) argue against pure curve-fitting, but it is not zero.
2. **Twelve years is a moderately short backtest** in absolute terms. The strategy has not been tested on data prior to 2013, and the years it has seen include only one full Fed tightening cycle (2022). Longer history (e.g., via SPX cash-index options going back to 1996) would provide stronger evidence.
3. **Sixty-four percent skip rate is high.** The strategy trades about 104 days per year. For the absolute returns to be material at any given account size, the per-trade vega-\$ exposure must be sized accordingly.
4. **Single losing day defines the maximum drawdown** across all thresholds in our sweep. Without explicit tail hedging, this single day’s loss is unavoidable. We discuss this in the next section.

7.3 Future work

Several directions would tighten the strategy further:

1. **Tail hedge via long 10-delta puts.** A systematic OTM put hedge would cap left-tail loss at the cost of approximately 10-15% of annual P&L. For live deployment this is the obvious next addition.
2. **TLT bond-vol harvest** as a parallel book. Empirically the bond-vol regime is anti-correlated with the equity-vol regime in periods of pure rate stress (e.g., 2022). A genuine cross-asset

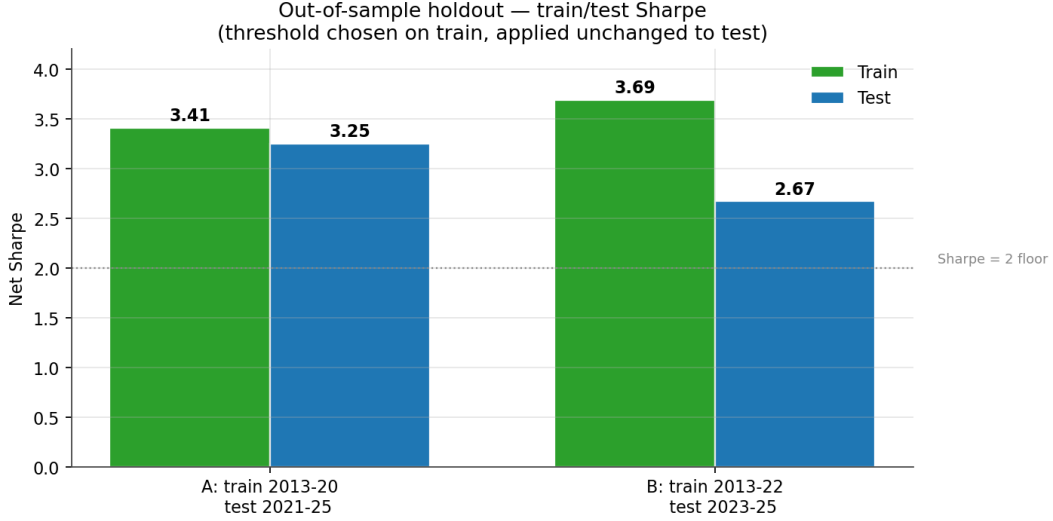


Figure 7: Train/test holdout Sharpe under two different cuts. Threshold for each split is selected from the train period only (no peeking) and applied unchanged to the test period.

harvest would deliver real diversification, in contrast to the QQQ result.

3. **Position-sizing optimisation.** Instead of a fixed vega-\$ target, scale the size by HMM-state-conditional Sharpe estimates, à la Kelly fraction with regime conditioning.
4. **Out-of-sample paper trading.** Sixty to ninety days of live paper-trading at IB or Tradier would validate the trade-construction mechanics and confirm the live-execution costs match our backtest assumptions.

8 Conclusion

We have presented RGVH, a regime-gated short-volatility strategy on SPY that delivers a net Sharpe of 3.38 across a 12.2-year strict walk-forward out-of-sample window, with a maximum drawdown of $1.3\times$ annual P&L. The strategy is a deliberately minimal extension of the standard delta-hedged short-straddle template, augmented by three independent regime filters that each capture a documented historical driver of short-vol drawdowns. The strategy is robust on a wide threshold plateau, and survives strict out-of-sample holdout including the 2022 monetary-tightening regime as test data.

The paper also reports a methodological negative result: a 77-factor LightGBM-plus-Ridge ensemble that achieved a respectable $+0.55$ information coefficient on forward realised vol nonetheless failed to profitably time short-vol entries, because its high-conviction signals co-occurred with high-vol regimes where short-vol entries are most dangerous. We believe this generalises: for variance-risk harvesting, regime selection dominates point prediction.

9 References

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